

# State density I: the one-dimensional standard integral at a general scale

Claude, with Daniel Murfet

2026-09-06

## Abstract

First unit of the state-density programme (the leading coefficient of `thm:TaylorTree` for general  $\xi, \eta$  as an integral of the fluctuation mass along the exceptional divisor). The one-dimensional standard integral  $I(c) = \int_0^b u^h \eta(u) e^{-\beta(cu^k)^2 + \beta cu^k \xi(u)} du$  is studied as a function of an arbitrary scale  $c \rightarrow \infty$ :  $c^p I(c) \rightarrow \frac{1}{k} A_{p-1}(\xi(0)) \eta(0)$  for data Lipschitz at 0 and without any sign condition on  $\eta(0)$ , and  $c^p |I(c)| \leq M e^{\beta L^2/2} \frac{1}{k} \int_0^\infty s^{p-1} e^{-(\beta/2)s^2} ds$  uniformly in  $c > 0$  when  $|\xi| \leq L, |\eta| \leq M$ . Zero `sorry/axiom`.

## 1 Setting

Along the divisor the inner one-dimensional integral has scale  $c = \sqrt{n} \prod_j v_j^{k_j}$  depending on the outer coordinates; the limit theorem is therefore needed in the scale variable, and dominated convergence in the outer coordinates needs a bound uniform in the scale.

## 2 The things formalised

```
noncomputable def oneDScale ( b c : ) ( h k : ) ( : → ) : :=
  u in Ioc 0 b, u ^ h * u * Real.exp ( - * ( c * u ^ k ) ^ 2 + * ( c * u ^ k ) * u )
theorem oneDScale_tendsto ... :
  Tendsto (fun c => c ^ (((h : ) + 1) / k) * oneDScale b c h k ) atTop
  ( (1 / (k : ) * weightedMass (((h : ) + 1) / k - 1) * ))
theorem oneDScale_abs_le ... (h L : u Ioc 0 b, | u | L)
  (h M : u Ioc 0 b, | u | M) :
  c ^ (((h : ) + 1) / k) * |oneDScale b c h k |
  M * Real.exp ( * L ^ 2 / 2 ) * (1 / (k : ))
  * weightedMass ( / 2) 0 (((h : ) + 1) / k - 1)
```

## 3 Proof strategy

The limit is unit 34's big- $O$  statement (which needs no  $\eta(0) \neq 0$ ) turned into a limit of  $n^{p/2} F(n)$  and composed with  $n = c^2$ ;  $\sqrt{c^2} = c$  and  $(c^2)^{p/2} = c^p$  for  $c > 0$ . The bound dominates the integrand pointwise by  $u^h M e^{\beta L^2/2} e^{-(\beta/2)(cu^k)^2}$  (unit 23's Gaussian inequality with  $t = (cu^k)^2$ ), then applies the chart power substitution and the scaling lemma of unit 44 to reduce to the Gaussian moment  $\int_0^\infty s^{p-1} e^{-(\beta/2)s^2}$ , a `weightedMass` at temperature  $\beta/2$  and  $a = 0$ .

## 4 Roadblocks and resolutions

- The Gaussian inequality is stated with  $\beta L\sqrt{t}$ ; matching  $\beta sL$  needed a `calc` step with `ring_nf` rather than `rwa`.
- `gcongr` on the final product left the inner inequality open; supply `mul_le_mul_of_nonneg_left` explicitly.

## 5 Outcome

Both inputs for integrating the one-dimensional theory along the divisor are in place; the next unit assembles them by Fubini and dominated convergence for a unique minimal coordinate.